

Assignment - 3

SQA

2022F-BCS-022

- Roll no: end with 22
 - ↳ is an even number and it lies b/w 1 and 99
 - ↳ replace line number 10 to 13 with "while loop"
 - ↳ replace line number 20 to 23 with "do-while loop"

Pseudo Code:

1: Read 22	14: Print B
2: $A = 22 * -1$	15: ELSE
3: $B = 22 * -10$	16: Print C
4: $C = 22 * 50$	17: END IF
5: $X = C + A - B$	18: ELSE
6: $Y = C + B - A$	19: IF $Y > Z$ THEN
7: $Z = A + B + C$	20: $I = 0$
8: IF $X > Y$ THEN	21:
9: IF $X > Z$ THEN	22:
10: $I = 0$	23:
11: WHILE $I < 10$ DO	24: Print B
12: Print A	25: ELSE
13: END WHILE	26: Print C
	27: END IF
	28: END IF

9) CFG

E-terminable

* Eliminating non-executable and non-decision statements:

550-228-75505

8: IF $X > Y$ THEN

9: IF $X > Z$ THEN

11: WHILE $I < 10$ DO

13: END WHILE

15: ELSE

17: ENDIF

18: ELSE

19: IF $Y > Z$ THEN

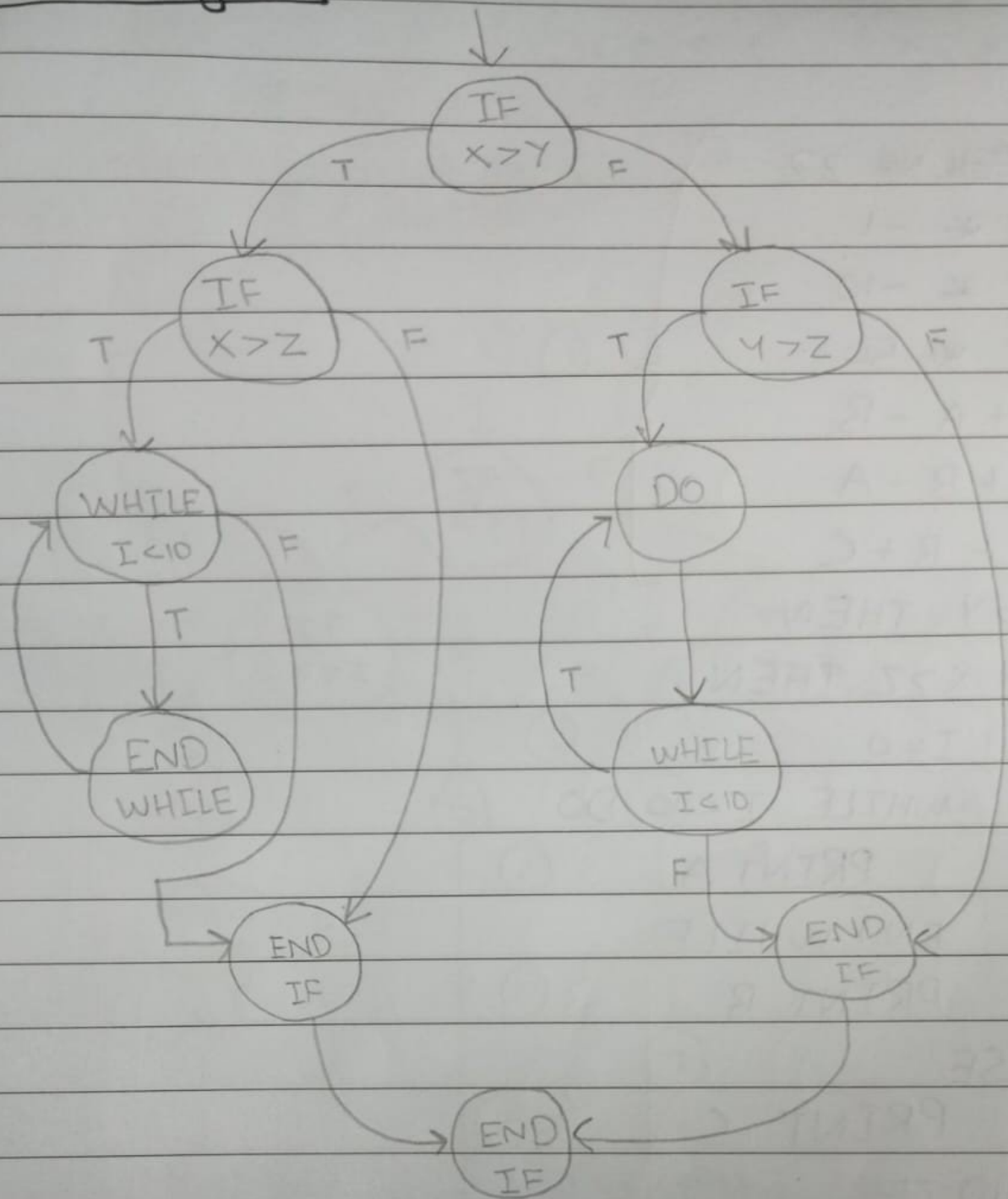
21: DO

23: WHILE $I < 10$

25: ELSE

27: END IF

28: END IF

Flow Diagram

Date: _____

b) HFG

1) READ ~~Roll No~~ 22

2) $A = 22 * -1$

3) $B = 22 * -10$

4) $C = 22 * 50$

5) $X = C + A - B$

6) $Y = C + B - A$

7) $Z = A + B + C$

8) IF $X > Y$ THEN

9) IF $X > Z$ THEN

10) $I = 0$

11) WHILE $I < 10$ DO

12) PRINT A

13) END WHILE

14) PRINT B

15) ELSE

16) PRINT C

17) END IF

18) ELSE

19) IF $Y > Z$ THEN

20) $I = 0$

21) DO

22) PRINT A

23) WHILE $I < 10$

24) PRINT B

①

②

③

④

⑤

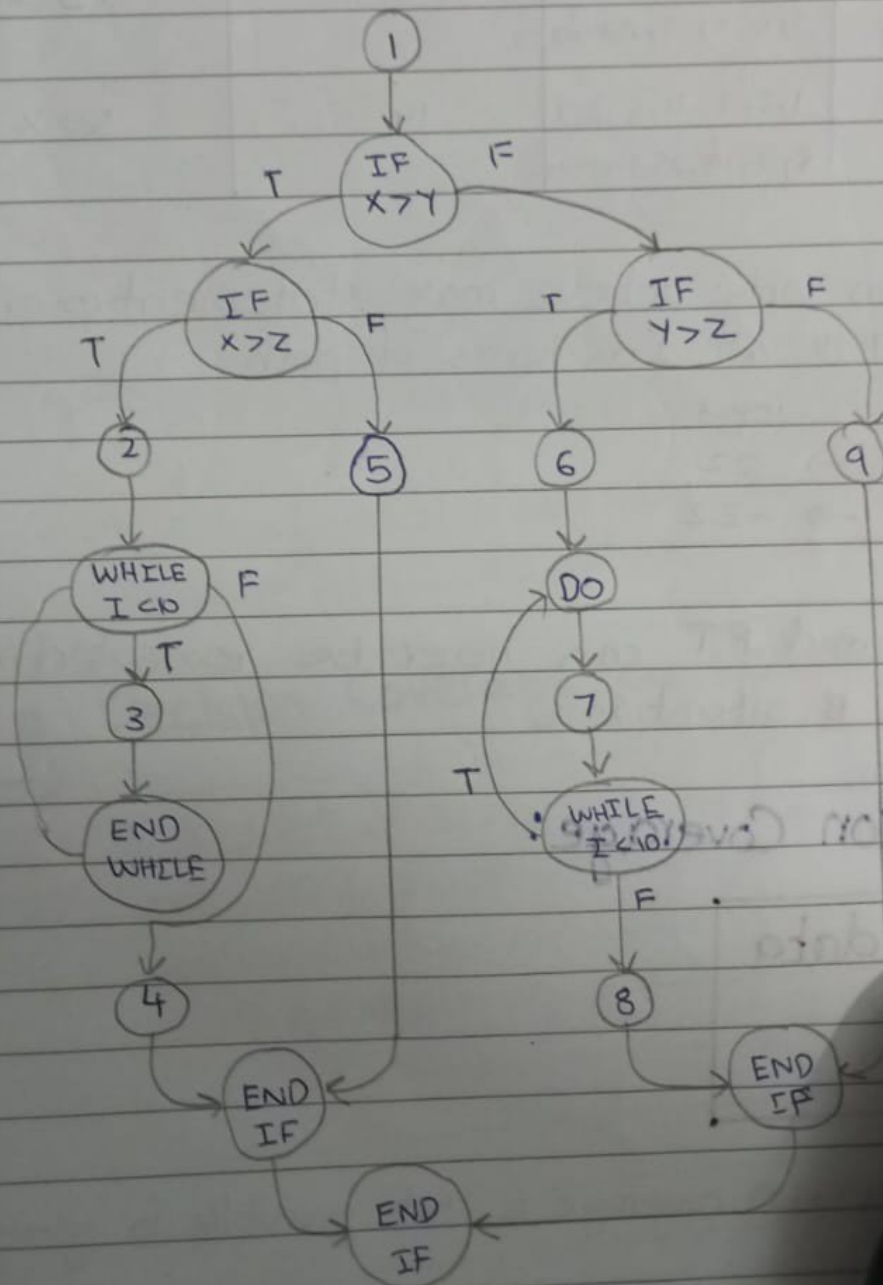
⑥

⑦

⑧

Date: _____

- 25) ELSE approved format 2 (5)
26) PRINT C 30 (9)
27) END IF start to all to all will will
28) END IF becomes becomes will



Date: _____

e) 100% statement coverage: 10/12 (ii)

We can reach maximum statement coverage of 75% using 2 test cases.

Achieving 100% statement coverage is logically impossible.

f) 100% decision coverage

Using '2' test cases we can achieve a maximum of 50% decision coverage.

Reaching 100% decision coverage in this scenario is not possible.

g) Decision Coverage by Roll no:

The last three digits of my roll number are "022", ~~to~~ using this^{as} an input we can achieve decision coverage of 25%. (Executing one of the four paths)

Date: _____

c) Statement Coverage

Value Roll no	line no. of stmts covered	No of stmts exercised	statement coverage
22	1, 2, 3, 4, 5, 6, 7, 8 9, 10, 11, 12, 13, 14, 28	16	57.14 %
-22	1, 2, 3, 4, 5, 6, 7 8, 18, 19, 25, 26, 27, 28	14	50 %

We can only achieve max statement coverage of 57.14 %. In terms of path.

input
TT → 22
FF → -22

TF and FT can never be executed in the given situation.

d) Decision Coverage:

Test data
22
-22

100% decision coverage is not possible in given scenario.

Date: _____

n) Statement coverage using Roll no. 10

Using my roll no we can achieve statement coverage of 57.14%. Following lines will not be executed:

$\Rightarrow$ 15, 16, 18, 19, 20, 21, 22, 23, 24, 25,
26, 27

Decision coverage 100% (?)
 $\longleftrightarrow$

Decision coverage by roll no: